Introduction To Flight Mechanics

Tutorial 9



source: flugrevue.de

Exercise: Trim and Linearization

a) Give a set of states that suffices to fully describe the rigid body motion of an aircraft. Classify the states as either “longitudinal” or “lateral”. Explain (by giving an example) why your particular choice of states is not the only possible solution.

b) State the primary control effectors of an aircraft. State the primary effect of each control (force or moment)?

c) Transform the well-known nonlinear differential equation of the spring-mass-damper system

$$m \cdot l^2 \cdot \ddot{\varphi}(t) = -m \cdot g \cdot l \cdot sin\, \varphi(t) - F_d\big(\dot{\varphi}(t)\big) + F(t) \cdot l \cdot \cos \varphi(t)$$

with the nonlinear damping force $F_d\big(\dot{\varphi}(t)\big) = d \cdot \dot{\varphi}(t) \cdot \cos\frac{\dot{\varphi}(t)}{\dot{\varphi}_{max}}$ to a linear system. Linearize around the steady-state operating point $\varphi(0) = \varphi_0$.

d) Linearize the equation for the y-position (which is decoupled from the longitudinal motion) and the equation for the bank angle around the condition: „steady-state straight flight without sideslip “.

$$0 = \dot{y} - V \cdot \sin\chi \cdot \cos\gamma = \dot{y} - V \cdot \sin(\Psi + \beta) \cdot \cos(\Theta - \alpha)$$

$$0 = \dot{\phi} - p - q \cdot \sin\Phi \cdot \tan\Theta - r \cdot \cos\Phi \cdot \tan\Theta$$

a)

I) Give a **complete set of states**, which **define the rigid body motion of the aircraft.**

II) Categorize the states according to lateral/longitudinal motion.

III) Explain (with an example) why the set of states is not the only possible solution.



a)

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What are the four categories of rigid body states?

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What are the four categories of rigid body states?

- Position
- Translation
- Attitude
- Rotation

Exercise 9a

Group	State	Symbol	Long / Lat
Position	x-Position Navigation frame	x_N	?
	y-Position Navigation frame	y_N	?
	z-Position Navigation frame	z_N	?
Alternative choice: ????????????????			
Translation	Absolute Kinematic velocity	V	?
	Angle of attack	α	?
	Angle of sideslip	β	?
Alternative choice: ?????? or ??????			

Group	State	Symbol	Long / Lat
Position Alternative choice: WGS-84 coordinates (λ, ϕ, h)	x-Position Navigation frame	x_N	-
	y-Position Navigation frame	y_N	-
	z-Position Navigation frame	z_N	-
Translation Alternative choice: (u_K, v_K, w_K) or (V, χ, γ)	Absolute Kinematic velocity	V	Longitudinal
	Angle of attack	α	Longitudinal
	Angle of sideslip	β	Lateral

Assuming horizontal flight in x_N → x_N = Long, y_N = Lat, z_N = Long

Exercise 9a

Group	State	Symbol	Long / Lat
Attitude	Bank angle	Φ	?
	Pitch angle	Θ	?
Alternative choice: quaternions (q_1, q_2, q_3, q_4)	Azimuth angle	Ψ	?
Rotation	Roll rate	p	?
	Pitch rate	q	?
	Yaw rate	r	?

Group	State	Symbol	Long / Lat
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	Pitch rate	q	Longitudinal
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b) What are the **primary controls of an aircraft**? What is the **primary effect of that control** (force or moment)?

Name	Symbol	Primary effect
Aileron deflection	ξ	
Elevator deflection	η	
Rudder deflection	ζ	
Thrust lever position	δ_T	



Force in x_B direction, roll, yaw and pitch moment.

Please assign these primary effects to the correct controls.

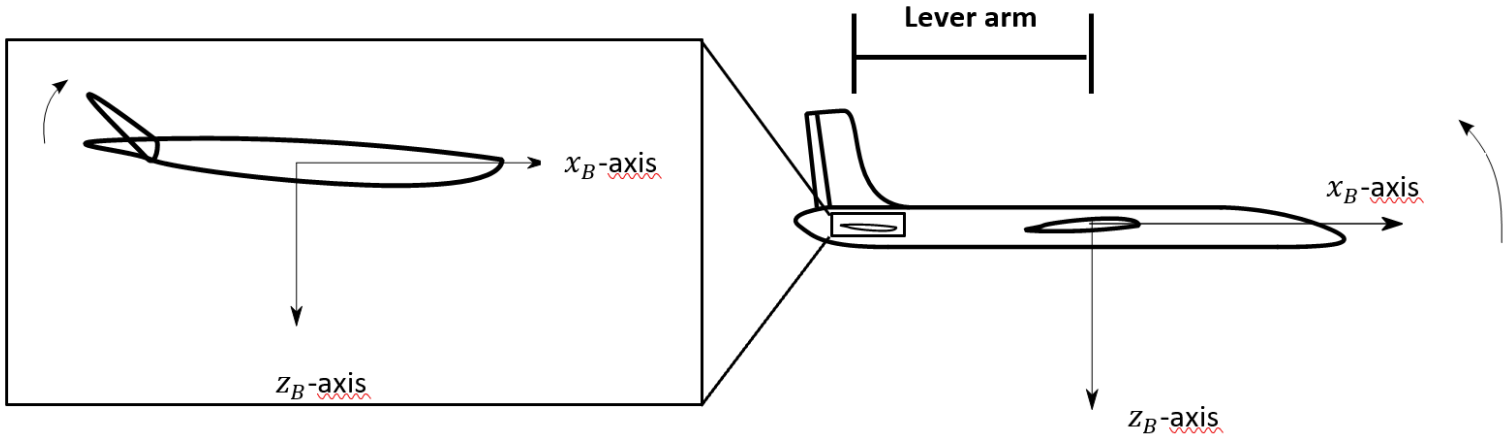
Name	Symbol	Primary effect
Aileron deflection	ξ	Roll moment
Elevator deflection	η	Pitch moment
Rudder deflection	ζ	Yaw moment
Thrust lever position	δ_T	Force in x_B direction

Definition of the FAA:

"Aircraft flight control systems consist of primary and secondary systems. The ailerons, elevator (or stabilisator), and rudder constitute the primary control system and are required to control an aircraft safely during flight. Wing flaps, leading edge devices, spoilers, and trim systems constitute the secondary control system and improve the performance characteristics of the airplane or relieve the pilot of excessive control forces"

Source: Federal Aviation Administration, Pilot's Handbook of Aeronautical Knowledge, Chapter 05, FAA-H-8083-25A, 2008

What‘s the effect of a pos/neg. elevator deflection?



Remark:
positive aileron/elevator/rudder deflection → negative roll/pitch/yaw moment

Is the depicted elevator deflection negative or positive?

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$$m \cdot l^2 \cdot \ddot{\varphi}(t) = -m \cdot g \cdot l \cdot sin\, \varphi(t) - F_d\big(\dot{\varphi}(t)\big) + F(t) \cdot l \cdot \cos \varphi(t)$$

with the nonlinear damping force

$$F_d\big(\dot{\varphi}(t)\big) = d \cdot \dot{\varphi}(t) \cdot \cos \frac{\dot{\varphi}(t)}{\dot{\varphi}_{max}}$$

to a linear system around the steady state operating point φ_0.

Linearizing Nonlinear Systems (1)

Motivation:

- For linear time invariant systems there are methods to analyze and design controllers.

- There is a proof, that for a globally exponentially stable linear system, the corresponding nonlinear system is locally exponentially stable (*Indirect Method of Lyapunov*) →*„acts as justification for working with locally linearized systems“*

Linearizing Nonlinear Systems (1)

Starting Point: Nonlinear State Space Model with Output Equation

$$\dot{x} = f(x,u) \qquad y = h(x,u) \qquad x(t_0) = x_s \qquad h(x_s,u_s) = y_s$$

Idea: Only **small** deviations of a stationary state x_s, for approximating the nonlinear function linearly in the proximity of the stationary state

$$x(t) = x_s + \delta x(t)$$

Function value

Value for the trim state

Deviation from the reference point

Reminder: The steady state must fulfill: $f(x_s,u_s) = 0$

Linearizing Nonlinear Systems (2)

Goal: linear state space model for the deviations of the steady state, input and outputs:

$$\begin{aligned}\delta\dot{\boldsymbol{x}} &= \boldsymbol{A}\cdot\delta\boldsymbol{x} + \boldsymbol{B}\cdot\delta\boldsymbol{u}\\ \delta\boldsymbol{y} &= \boldsymbol{C}\cdot\delta\boldsymbol{x} + \boldsymbol{D}\cdot\delta\boldsymbol{u}\end{aligned} \quad \Longleftarrow \quad \dot{x} = f(x,u) \qquad y = h(x,u)$$

Idea: Taylor approximation of $f(x,u)$ and $h(x,u)$ around (x_s, u_s), truncated after the linear term:

$$f(x,u) \approx \cancel{f(x_s,u_s)}^{\,0} + \left.\frac{\partial f(x,u)}{\partial x}\right|_{\substack{x=x_s\\u=u_s}} (x-x_s) + \left.\frac{\partial f(x,u)}{\partial u}\right|_{\substack{x=x_s\\u=u_s}} (u-u_s)$$

$$\delta\dot{\boldsymbol{x}} = \boldsymbol{A}\cdot\delta\boldsymbol{x} + \boldsymbol{B}\cdot\delta\boldsymbol{u}$$

$$h(x,u) = \boldsymbol{y}_s + \delta\boldsymbol{y} \approx h(x_s,u_s) + \left.\frac{\partial h(x,u)}{\partial x}\right|_{\substack{x=x_s\\u=u_s}} (x-x_s) + \left.\frac{\partial h(x,u)}{\partial u}\right|_{\substack{x=x_s\\u=u_s}} (u-u_s)$$

$$\cancel{\boldsymbol{y}_s} + \delta\boldsymbol{y} = \cancel{\boldsymbol{y}_s} + \boldsymbol{C}\cdot\delta\boldsymbol{x} + \boldsymbol{D}\cdot\delta\boldsymbol{u}$$

Linearizing Nonlinear Systems (3)

Result: Linear, time invariant state space model:

$$\delta\dot{\boldsymbol{x}} = \boldsymbol{A} \cdot \delta\boldsymbol{x} + \boldsymbol{B} \cdot \delta\boldsymbol{u}$$
$$\delta\boldsymbol{y} = \boldsymbol{C} \cdot \delta\boldsymbol{x} + \boldsymbol{D} \cdot \delta\boldsymbol{u}$$



Do you know the nomencature? Fill in the names

with

$\boldsymbol{x}(t) \in \mathbb{R}^n$	? vector
$\boldsymbol{y}(t) \in \mathbb{R}^r$	? vector
$\boldsymbol{u}(t) \in \mathbb{R}^m$	? vector
$\boldsymbol{A} \in \mathbb{R}^{n \times n}$	? matrix
$\boldsymbol{B} \in \mathbb{R}^{n \times m}$	? matrix
$\boldsymbol{C} \in \mathbb{R}^{r \times n}$	? matrix
$\boldsymbol{D} \in \mathbb{R}^{r \times m}$	? matrix

Remark: For simplifying the equations, the „δ“ is often omitted. It is important to note this difference, (especially for designing controllers which are build upon linear models), because the trim points have to be considered!

Linearizing Nonlinear Systems (3)

Result: Linear, time invariant state space model:

$$\delta\dot{\boldsymbol{x}} = \boldsymbol{A} \cdot \delta\boldsymbol{x} + \boldsymbol{B} \cdot \delta\boldsymbol{u}$$
$$\delta\boldsymbol{y} = \boldsymbol{C} \cdot \delta\boldsymbol{x} + \boldsymbol{D} \cdot \delta\boldsymbol{u}$$

with

$\boldsymbol{x}(t) \in \mathbb{R}^n$	state vector
$\boldsymbol{y}(t) \in \mathbb{R}^r$	output vector
$\boldsymbol{u}(t) \in \mathbb{R}^m$	input vector
$\boldsymbol{A} \in \mathbb{R}^{n \times n}$	system matrix
$\boldsymbol{B} \in \mathbb{R}^{n \times m}$	input matrix
$\boldsymbol{C} \in \mathbb{R}^{r \times n}$	output matrix
$\boldsymbol{D} \in \mathbb{R}^{r \times m}$	feedthrough matrix

Remark: For simplifying the equations, the „δ“ is often omitted. It is important to note this difference, (especially for designing controllers which are build upon linear models), because the trim points have to be considered!

Nonlinear dynamics

Damping force $F_d(\dot{\varphi}(t))$

$$m \cdot l^2 \cdot \ddot{\varphi}(t) = -m \cdot g \cdot l \cdot sin\, \varphi(t) - d \cdot \dot{\varphi}(t) \cdot \cos\frac{\dot{\varphi}(t)}{\dot{\varphi}_{max}} + F(t) \cdot l \cdot \cos \varphi(t)$$

Step 1 → Identify states and inputs

State: $\varphi(t)$

Input: $u = F(t)$

Problem: higher order system (linear SS-model is a first-order system)

Step 2 → Reduce system order:

$x_1 = \varphi$

$x_2 = \dot{\varphi}$

$u = F(t)$

$$\dot{x}_1 = x_2 = \dot{\varphi}$$

$$m \cdot l^2 \cdot \dot{x}_2(t) = -m \cdot g \cdot l \cdot sin\, x_1(t) - d \cdot x_2(t) \cdot \cos\frac{x_2(t)}{x_{2,max}} + u(t) \cdot l \cdot \cos x_1(t)$$

$$m \cdot l^2 \cdot \dot{x}_2 = -m \cdot g \cdot l \cdot sin\, x_1 - d \cdot x_2 \cdot \cos\frac{x_2}{x_{2,max}} + u \cdot l \cdot \cos x_1$$

State derivatives: $\dot{x}_1 = x_2 = f_1(\boldsymbol{x}, u)$

State Vector $\boldsymbol{x}$ contains x_1 and x_2
Therefore $\dot{\boldsymbol{x}}$ contains $\dot{x}_1$ which is x_2

$$\dot{x}_2 = f_2(\boldsymbol{x}, \dot{\boldsymbol{x}}, u) = -\frac{d \cdot x_2 \cos\frac{x_2}{x_{2,max}}}{ml^2} - g \cdot \frac{\sin x_1}{l} + u \cdot \frac{\cos x_1}{ml}$$

Problem: first derivative of x_2: $\dot{x}_2 = \ddot{\varphi}$ is nonlinear.

Step 3→ Linearization around operating condition $\boldsymbol{x}_{\mathrm{ref}}$

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, u) = \begin{bmatrix} f_1(\boldsymbol{x}, u) \\ f_2(\boldsymbol{x}, u) \end{bmatrix}$$

around $\boldsymbol{x}_{\mathrm{ref}}$

$$\dot{\boldsymbol{x}} = \boldsymbol{A} \cdot \boldsymbol{x} + \boldsymbol{B} \cdot \boldsymbol{u}$$

I. Reference point/ steady state operation point

At steady state point $\boldsymbol{x}_{\mathrm{ref}}$: $\dot{x}_{1,\mathrm{ref}} = \dot{x}_{2,\mathrm{ref}} = 0$

Reference states: $x_{1,\mathrm{ref}} = \varphi_0$

$$x_{2,\mathrm{ref}} = \dot{x}_{1,\mathrm{ref}} = 0$$

Compute input at reference condition:

$$\dot{x}_{2,\mathrm{ref}} = f_2(\boldsymbol{x}_{\mathrm{ref}}, \dot{\boldsymbol{x}}_{\mathrm{ref}}, u_{\mathrm{ref}}) = -\frac{d \cdot x_{2,\mathrm{ref}} \cos\frac{x_{2,\mathrm{ref}}}{x_{2,max}}}{ml^2} - g \cdot \frac{\sin x_{1,\mathrm{ref}}}{l} + u_{\mathrm{ref}} \cdot \frac{\cos x_{1,\mathrm{ref}}}{ml}$$

$$= 0 - g \cdot \frac{\sin \varphi_0}{l} + u_{\mathrm{ref}} \cdot \frac{\cos \varphi_0}{ml} = 0.$$

$$\Rightarrow \quad u_{ref} = mg \cdot \tan \varphi_0$$

II. Linearized System/Input Dynamics

General: $$\dot{\boldsymbol{x}} = \boldsymbol{A} \cdot \boldsymbol{x} + \boldsymbol{B} \cdot \boldsymbol{u}$$

System matrix $$\boldsymbol{A} = \left.\frac{\partial \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})}{\partial \boldsymbol{x}}\right|_{\substack{\boldsymbol{x}=\boldsymbol{x}_{ref} \\ \boldsymbol{u}=\boldsymbol{u}_{ref}}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

Input matrix $$\boldsymbol{B} = \left.\frac{\partial \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})}{\partial \boldsymbol{u}}\right|_{\substack{\boldsymbol{x}=\boldsymbol{x}_{ref} \\ \boldsymbol{u}=\boldsymbol{u}_{ref}}} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \cdots & \frac{\partial f_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \cdots & \frac{\partial f_n}{\partial u_m} \end{bmatrix}$$

$$f_1(\boldsymbol{x}, u) = \dot{x}_1 = x_2$$

$$f_2(\boldsymbol{x}, \dot{\boldsymbol{x}}, u) = \dot{x}_2 = -\frac{dx_2 \cos\frac{x_2}{x_{2,max}}}{ml^2} - g \cdot \frac{\sin x_1}{l} + u \cdot \frac{\cos x_1}{ml}$$

A-Matrix:

Derivatives of f_1:

$$\left.\frac{\partial f_1}{\partial x_1}\right|_{ref} = 0$$

$$\left.\frac{\partial f_1}{\partial x_2}\right|_{ref} = 1$$

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ ? & ? \end{bmatrix}$$

$$f_1(\boldsymbol{x}, u) = \dot{x}_1 = x_2$$

$$f_2(\boldsymbol{x}, \dot{\boldsymbol{x}}, u) = \dot{x}_2 = -\frac{d x_2 \cos\frac{x_2}{x_{2,max}}}{m l^2} - g \cdot \frac{\sin x_1}{l} + u \cdot \frac{\cos x_1}{m l}$$

Derivative of f_2 w.r.t. x_1:

$$\left.\frac{\partial f_2}{\partial x_1}\right|_{ref} = 0 - \frac{g}{l}\cos x_{1,ref} - \frac{u_{ref}}{ml}\sin x_{1,ref} \qquad u_{ref} = mg \cdot \tan\varphi_0$$

$$= -\frac{g}{l}\cos x_{1,ref} - \frac{mg \cdot \tan x_{1,ref}}{ml}\sin x_{1,ref}$$

$$= -\frac{g}{l}\cos x_{1,ref} - \frac{\frac{g}{l}\sin^2 x_{1,ref}}{\cos x_{1,ref}} = -\frac{\frac{g}{l}\cos^2 x_{1,ref}}{\cos x_{1,ref}} - \frac{\frac{g}{l}\sin^2 x_{1,ref}}{\cos x_{1,ref}}$$

$$= -\frac{g}{l}\frac{1}{\cos x_{1,ref}}\left(\cos^2 x_{1,ref} + \sin^2 x_{1,ref}\right) = -\frac{g}{l}\frac{1}{\cos x_{1,ref}}$$

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l}\frac{1}{\cos x_{1,ref}} & ? \end{bmatrix}$$

$$f_1(\boldsymbol{x}, u) = \dot{x}_1 = x_2$$

$$f_2(\boldsymbol{x}, \dot{\boldsymbol{x}}, u) = \dot{x}_2 = -\frac{d x_2 \cos\frac{x_2}{x_{2,max}}}{ml^2} - g \cdot \frac{\sin x_1}{l} + u \cdot \frac{\cos x_1}{ml}$$

Derivative of f_2 w.r.t. x_2:

$$\left.\frac{\partial f_2}{\partial x_2}\right|_{ref} = -\frac{d \cdot \cos\frac{x_{2,ref}}{x_{2,max}}}{ml^2} + \frac{d \cdot x_{2,\mathrm{ref}} \cdot \sin\frac{x_{2,ref}}{x_{2,max}}}{ml^2 \cdot x_{2,max}}$$

$$= -\frac{d}{ml^2}$$

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l}\frac{1}{\cos x_{1,ref}} & -\frac{d}{ml^2} \end{bmatrix}$$

Reference state:

$\dot{x}_{1,ref} = \dot{x}_{2,ref} = x_{2,ref} = 0$

$$f_1(\boldsymbol{x}, u) = \dot{x}_1 = x_2$$

$$f_2(\boldsymbol{x}, \dot{\boldsymbol{x}}, u) = \dot{x}_2 = -\frac{dx_2 \cos\frac{x_2}{x_{2,max}}}{ml^2} - g \cdot \frac{\sin x_1}{l} + u \cdot \frac{\cos x_1}{ml}$$

B-Matrix:

Derivatives of f_1 and f_2 w.r.t. u:

$$\left.\frac{\partial f_1}{\partial u_1}\right|_{ref} = 0$$

$$\left.\frac{\partial f_2}{\partial u_1}\right|_{ref} = \frac{\cos x_{1,ref}}{ml}$$

$$\boldsymbol{B} = \begin{bmatrix} 0 \\ \frac{\cos x_{1,ref}}{ml} \end{bmatrix}$$

Assembling results:

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l}\frac{1}{\cos x_{1,ref}} & -\frac{d}{ml^2} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ \frac{\cos x_{1,ref}}{ml} \end{bmatrix}$$

Always recall that we only consider incremental changes from the reference condition!

$$x_1 - x_{1,ref}$$
$$x_2 - x_{2,ref}$$
$$u_1 - u_{1,ref}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l}\frac{1}{\cos x_{1,ref}} & -\frac{d}{ml^2} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{\cos x_{1,ref}}{ml} \end{bmatrix} \cdot u_1$$

$$\dot{\boldsymbol{x}} = \boldsymbol{A} \cdot \boldsymbol{x} + \boldsymbol{B} \cdot \boldsymbol{u}$$

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$$0 = \dot{\phi} - p - q \cdot \sin\Phi \cdot \tan\Theta - r \cdot \cos\Phi \cdot \tan\Theta$$

d) Linearize the y position equation (**decoupled from longitudinal motion**) and the bank angle equation around the **reference point „stationary straight flight without sideslip“**.

Lateral position differential equation

$$0 = \dot{y} - V \cdot \sin\chi \cdot \cos\gamma = \dot{y} - V \cdot \sin(\Psi + \beta) \cdot \cos(\Theta - \alpha)$$

Bank angle differential equation

$$0 = \dot{\phi} - p - q \cdot \sin\Phi \cdot \tan\Theta - r \cdot \cos\Phi \cdot \tan\Theta$$

Lateral Motion States = β, p, r, ϕ, Ψ

Reference point „stationary straight flight without sideslip“.

$\beta_0 = ?$

$\Phi_0 = ?$

$p = q = r = ?$



Fill the gaps/question marks

Decoupled from the longitudinal motion

→ Change in longitudinal motion states is zero!
We therefore only consider the lateral states: $(\beta, p, r, \phi, \Psi)$

Lateral position differential equation

$$0 = \dot{y} - V \cdot \sin\chi \cdot \cos\gamma = \dot{y} - V \cdot \sin(\Psi + \beta) \cdot \cos(\Theta - \alpha)$$

χ γ

This only holds for the stationary straight flight!

Lateral Motion States = β, p, r, ϕ, Ψ

Reference point „stationary straight flight without sideslip“.

$$\beta_0 = 0$$

$$\Phi_0 = 0$$

$$p = q = r = 0$$

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χ γ

This only holds for the stationary straight flight!

$$0 = \dot{y} - V \cdot \sin(\Psi + \beta) \cdot \cos(\Theta - \alpha)$$

$$\beta_0 = 0 \qquad \Phi_0 = 0 \qquad p = q = r = 0$$

explicit $\quad f(\boldsymbol{x}) = \dot{y} = V \cdot \sin(\Psi + \beta) \cdot \cos(\Theta - \alpha)$

$$(\beta, p, r, \phi, \Psi)$$

$$\left.\frac{\partial f}{\partial p}\right|_0 = \left.\frac{\partial f}{\partial r}\right|_0 = \left.\frac{\partial f}{\partial \phi}\right|_0 = \left.\frac{\partial f}{\partial u}\right|_0 = 0$$

$$\left.\frac{\partial f}{\partial \beta}\right|_0 = V_0 \cdot \cos(\Psi_0 + \beta_0)\cos(\Theta_0 - \alpha_0) \qquad \left.\frac{\partial f}{\partial \Psi}\right|_0 = V_0 \cdot \cos(\Psi_0 + \beta_0)\cos(\Theta_0 - \alpha_0)$$

We know that $\beta_0 = 0$.

$$\left.\frac{\partial f}{\partial \beta}\right|_0 = V_0 \cdot \cos(\Psi_0)\cos(\Theta_0 - \alpha_0) \qquad \left.\frac{\partial f}{\partial \Psi}\right|_0 = V_0 \cdot \cos(\Psi_0)\cos(\Theta_0 - \alpha_0)$$

$$\delta\dot{y} = \begin{bmatrix} \left.\frac{\partial f}{\partial \beta}\right|_0 & \left.\frac{\partial f}{\partial p}\right|_0 & \left.\frac{\partial f}{\partial r}\right|_0 & \left.\frac{\partial f}{\partial \phi}\right|_0 & \left.\frac{\partial f}{\partial \Psi}\right|_0 \end{bmatrix} \cdot \begin{bmatrix} \delta\beta \\ \delta p \\ \delta r \\ \delta\Phi \\ \delta\Psi \end{bmatrix}$$

$$= \begin{bmatrix} V_0 \cdot \cos(\Psi_0)\cos(\Theta_0 - \alpha_0) & 0 & 0 & 0 & \left. V_0 \cdot \cos(\Psi_0)\cos(\Theta_0 - \alpha_0)\right|_0 \end{bmatrix} \cdot \begin{bmatrix} \delta\beta \\ \delta p \\ \delta r \\ \delta\Phi \\ \delta\Psi \end{bmatrix}$$

$$\delta\dot{y} = Cx + Du$$

$$= \left[V_0 \cdot \cos(\Psi_0)\cos(\Theta_0 - \alpha_0) \quad 0 \quad 0 \quad 0 \quad V_0 \cdot \cos(\Psi_0)\cos(\Theta_0 - \alpha_0)\Big|_0 \right] \cdot \begin{bmatrix} \delta\beta \\ \delta p \\ \delta r \\ \delta\Phi \\ \delta\Psi \end{bmatrix}$$

$$\boxed{\delta\dot{y} = V_0 \cdot \cos(\Psi_0)\cos(\Theta_0 - \alpha_0)\,\delta\beta + V_0 \cdot \cos(\Psi_0)\cos(\Theta_0 - \alpha_0)\,\delta\Psi}$$

Bank angle differential equation:

$$0 = \dot{\phi} - p - q \cdot \sin\Phi \cdot \tan\Theta - r \cdot \cos\Phi \cdot \tan\Theta$$

$$\beta_0 = 0$$
$$p_0 = q_0 = r_0 = 0$$
$$\Phi_0 = 0$$

explicit $f(\boldsymbol{x}) = \dot{\Phi} = p + q \sin\phi \cdot \tan\Theta + r \cdot \cos\phi \cdot \tan\Theta$

Θ and q → Longitudinal motion, no derivatives required!

$(\beta, p, r, \phi, \Psi)$ → *Lateral motion*

$$\left.\frac{\partial f}{\partial \beta}\right|_0 = \left.\frac{\partial f}{\partial \Psi}\right|_0 = \left.\frac{\partial f}{\partial u}\right|_0 = 0$$

$$\left.\frac{\partial f}{\partial \Phi}\right|_0 = 0 + q_0 \cdot \cos\Phi_0 \cdot \tan\Theta_0 - r_0 \cdot \sin\Phi_0 \cdot \tan\Theta_0 = 0$$

(q_0: 0, r_0: 0)

$$\left.\frac{\partial f}{\partial p}\right|_0 = 1$$

$$\left.\frac{\partial f}{\partial r}\right|_0 = \cos\phi_0 \cdot \tan\Theta_0$$

($\cos\phi_0$: 1, $\phi_0 = 0$)

$$\delta\dot{\Phi} = C\delta x + D\delta u = [0 \quad 1 \quad \tan\Theta_0 \quad 0 \quad 0] \cdot \begin{bmatrix} \delta\beta \\ \delta p \\ \delta r \\ \delta\Phi \\ \delta\Psi \end{bmatrix}$$

$$\boxed{\delta\dot{\Phi} = 1 \cdot \delta p + \tan\Theta_0 \cdot \delta r}$$